11^{th} Set of Problems

1. For a certain *RR manipulator*, such as the one illustrated in Figure 1(a), the equations of motion are given by

$$\begin{pmatrix} 4+c_2 & 1+c_2 \\ 1+c_2 & 1 \end{pmatrix} \begin{pmatrix} \ddot{\vartheta}_1 \\ \ddot{\vartheta}_2 \end{pmatrix} + \begin{pmatrix} -s_2\left(\dot{\vartheta}_2^2 + 2\,\dot{\vartheta}_1\,\dot{\vartheta}_2\right) \\ s_2\,\dot{\vartheta}_1^2 \end{pmatrix} = \begin{pmatrix} \tau_1 \\ \tau_2 \end{pmatrix}.$$

Assume that both joints are free to move, and that the manipulator is controlled by an inverse dynamics controller.

(a) Design the controller that will provide a closed-loop frequency of $10\,\mathrm{rad/s}$ on the first joint and $20\,\mathrm{rad/s}$ on the second joint, and be critically damped over the entire workspace.

(b) By means of a block diagram, illustrate the previous inverse dynamics controller.

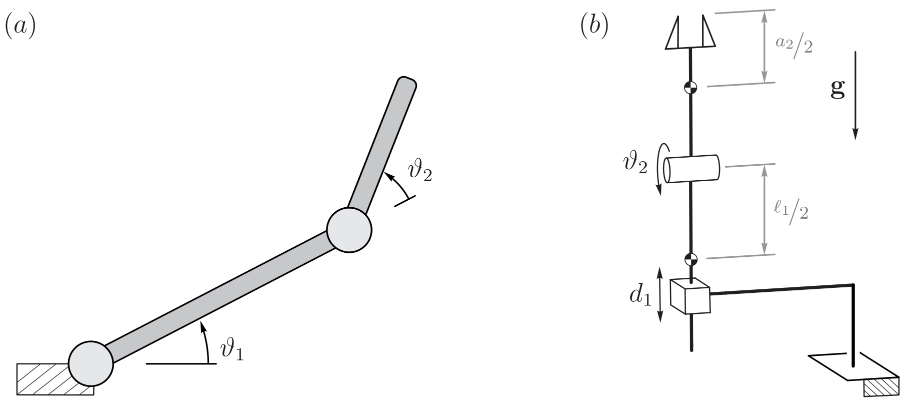

Figure 1: (a) Two-link planar arm. (b) Stadium manipulator.

2. Consider a prismatic-revolute manipulator, such as the one illustrated in Figure 1(b), whose equations of motion have been written as

$$\begin{pmatrix} m_1+m_2 & m_2\,\frac{a_2}{2}\,c_2 \\ m_2\,\frac{a_2}{2}\,c_2 & I_{zz}^{(2)} + m_2\,\frac{a_2^2}{4} \end{pmatrix} \begin{pmatrix} \ddot{d}_1 \\ \ddot{\vartheta}_2 \end{pmatrix} + \begin{pmatrix} -m_2\,\frac{a_2}{2}\,s_2\,\dot{\vartheta}_2 \\ 0 \end{pmatrix} + \begin{pmatrix} (m_1+m_2)\,|\mathrm{g}| \\ m_2\,|\mathrm{g}|\,\frac{a_2}{2}\,c_2 \end{pmatrix} = \boldsymbol{\tau}\,.$$

Design a controller through inverse dynamics, in order to obtain a closed loop bandwidth of ω_{n_1} and ω_{n_2} $\mathrm{rad/s}$ for the joints, respectively, and critical damping.